

Tintin Wu (Chun-Tin Wu)

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SUMMARY

AI Software Engineer with 3+ years of experience building production-grade AI systems across agentic AI, multi-modal LLM/VLM, and computer vision. Currently developing multi-agent orchestration systems at Binance using Model Context Protocol (MCP), Agent Development Kit (ADK), and Agent2Agent (A2A) protocols. Published researcher in 3D/4D scene rendering (IEEE ICIP) and AI agent tool calling (EACL Findings), with a track record of leading technical teams and shipping AI solutions from research to deployment.

SKILLS

- **Programming:** Python, C#, CUDA
- **AI / ML Frameworks:** PyTorch, Hugging Face, vLLM, OpenCV, Scikit-learn
- **Agentic AI:** Model Context Protocol (MCP), Agent Development Kit (ADK), Agent2Agent (A2A), LangGraph, LangSmith, AutoGen, NVIDIA NeMo Agent Toolkit, Opik
- **Domains:** Multi-agent orchestration, LLM/VLM training and deployment, 3D/4D Gaussian Splatting, novel view synthesis, face recognition and anti-spoofing
- **Other Tools:** Unity, Android Studio, Git

WORK EXPERIENCE

AI Software Engineer — Binance

Sep 2025 – Present

- Built InfraBot, a universal multi-agent orchestration system that unifies HTTP/HTTPS APIs, Agent2Agent (A2A) endpoints, and internal chatroom platforms, enabling previously siloed services to interoperate through a single agent network.
- Designed the sub-agent architecture and orchestration layer for a multi-modal agentic AI system, by building on Agent Development Kit (ADK), A2A, LangGraph, and LangSmith.
- Launched Agent-Runtime, an internal agent factory that turns AI agent deployment from manual setup into an instant, self-service process.
- Made OpenClaw agent runs traceable end-to-end, by introducing Opik tracing with a span-tree structure that binds main agents, sub-agents, and tool calls in every run.

Associate Researcher, Agentic AI Team Leader — Delta Electronics

Nov 2024 – Sep 2025

- Led the Agentic AI team to design and ship a Model Context Protocol (MCP)-based agentic AI ecosystem adopted within Delta Electronics, by architecting the system on MCP, ADK, A2A, NVIDIA NeMo Agent Toolkit, AutoGen, and LangGraph.
- Trained and deployed large language models (LLM) and vision-language models (VLM) into production, using Python, PyTorch, Hugging Face, and vLLM.
- Automated office workflows company-wide, by delivering LLM-agent-powered automation built on the team's agentic AI ecosystem.

Research Assistant — Research Center for Information Technology Innovation, Academia Sinica

Jul 2024 – Oct 2024

- Achieved state-of-the-art dynamic scene rendering performance with lower memory consumption and faster training/inference than prior methods, by developing 4D Neural Voxel Splatting, which encodes 3D Gaussians with neural voxels and reinforces Gaussian deformation with K-Plane modules; published at IEEE ICIP 2026.
- Delivered photorealistic novel view synthesis with real-time rendering, rapid convergence, and low computational cost on dynamic scenes, using Python, PyTorch, and custom CUDA kernels.
- Reconstructed surfaces from dynamic scenes, by introducing Signed Distance Functions (SDF) into 4D Gaussian Splatting, combining Gaussian-rendered depth maps with ray tracing, and applying mutual supervision between RGB and SDF branches in a dual-branch architecture.

- Extracted 3D geometry and meshes from static scenes, by training a dual-branch SDF-head/RGB-head architecture on top of 3D Gaussian Splatting.

Machine Learning Engineer — Taiwan Mobile

Sep 2022 – Jul 2023

- Delivered face and ID-card recognition models for the company's eKYC (electronic Know Your Customer) product, by applying computer vision and deep learning methods to identity verification.
- Achieved 99.97% accuracy on the CelebA-Spoof dataset for face presentation attack detection, by ensembling SwinV2, EfficientNet, and RegNet backbones with a multi-head classifier and cross-validation.
- Detected fake ID cards while training on genuine samples only, by building a one-class detection pipeline that generates synthetic fakes with the Segment Anything Model (SAM) through face mix-matching, combined with cross-validated model ensembles to prevent overfitting.
- Verified anti-counterfeiting holograms on national ID cards from video input, by analyzing hue distributions across frames with guided capture instructions and threshold-based classification (Python, OpenCV).

EDUCATION

M.S., Communication Engineering — National Taiwan University | Mar 2023 – Jul 2024

Graduate Student Researcher, Research Center for Information Technology Innovation, Academia Sinica (advisor: Prof. Jun-Cheng Chen). Coursework: Artificial Intelligence, Machine Learning, Generative AI, Trustworthy AI.

B.S., Electrical Engineering — National Taiwan University | Sep 2018 – Jun 2022

Undergraduate Research Fellow, Vision and Learning Lab (advisor: Prof. Yu-Chiang Frank Wang). Coursework: Computer Vision, Deep Learning.

PUBLICATIONS

- Chun-Tin Wu, Jun-Cheng Chen. "4D Neural Voxel Splatting: Dynamic Scene Rendering with Voxelized Gaussian Splatting." *IEEE International Conference on Image Processing (ICIP)*, 2026. arXiv:2511.00560
- Jia-Kai Dong, I-Wei Huang, Chun-Tin Wu, Yi-Tien Tsai. "MSC-Bench: A Rigorous Benchmark for Multi-Server Tool Orchestration." *Findings of the Association for Computational Linguistics (EACL)*, 2026.

SELECTED PROJECTS

- **Semantic Scene Completion** (Python, PyTorch): Improved semantic scene completion on 3D outdoor point clouds (SemanticKITTI), by designing a Scene-Instance-Scene pipeline that combines a compact scene-completion model for initialization, a panoptic segmentation model for instance proposals, and lightweight per-instance completion models to overcome dataset inconsistencies.
- **Lightweight Face Detection** (Python, PyTorch): Achieved 80.82 precision and 6.25 NME with a model under 5 MB, by combining inverted residuals, Feature Pyramid Network (FPN), Single Stage Headless detector (SSH), and Deformable Convolution Network (DCN).
- **Face Anti-Spoofing** (Python, PyTorch): Achieved 99.3% accuracy on the Oulu-NPU and SiW datasets, by pairing a MobileNetV2 backbone with a bidirectional LSTM over 11-frame video sequences.
- **Peng-Hu VR Project** (Unity, C#, Android Studio): Shipped a 15-minute VR first-person shooter set in the Peng-Hu islands on Google Cardboard, by building 3D models of the islands and terrain features within tight mobile-hardware constraints.

ADDITIONAL

- **Languages:** Mandarin Chinese (native), English (fluent, TOEIC 960)
- **Leadership:** Leader, Martial Arts School Team, National Taiwan University